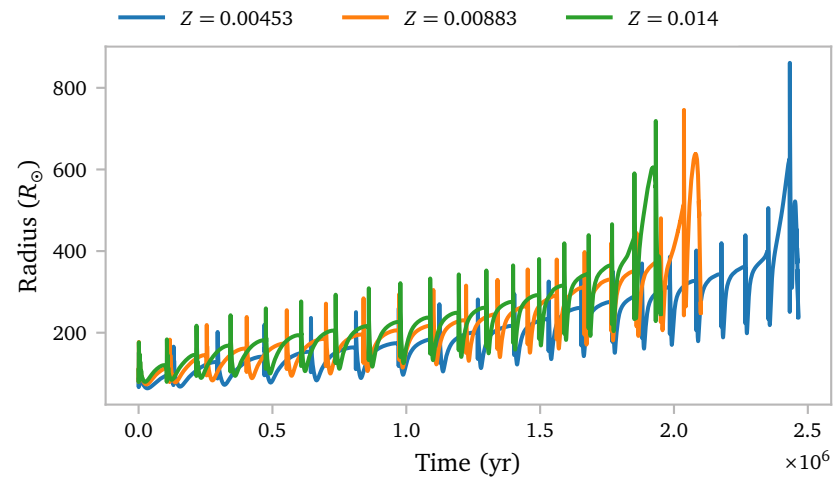


Notes Week 21

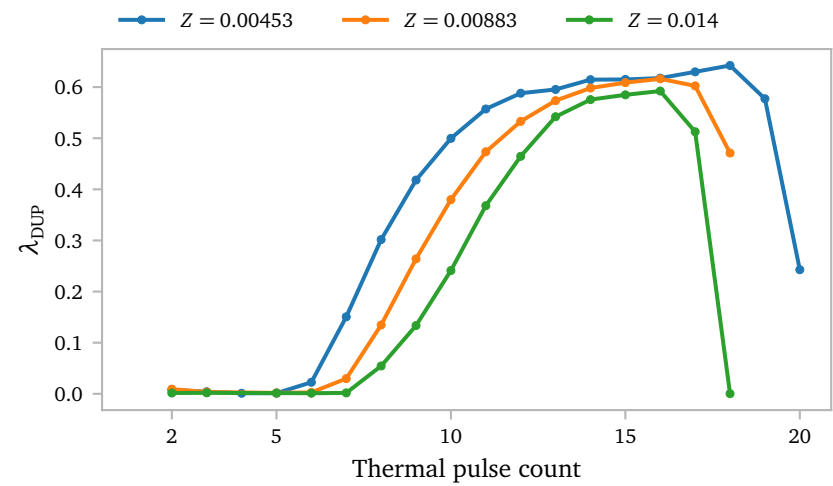
a. Dredge-up and Abundances

To start, I want to plot the radius evolution for the models.



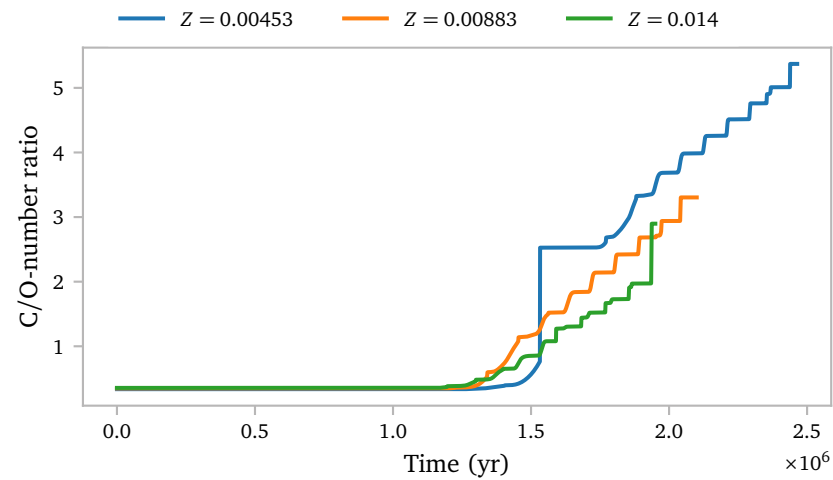
All seem fine, and the general trends that I saw last week seem to hold for the lower metallicity star as well.

Now, more interestingly, let me plot the dredge-up efficiencies.



It appears that dredge is slightly more efficient for lower metallicity stars. This is in line with literature predictions (Herwig, 2005; Karakas et al., 2002). Metallicity also seems to have an inverse relation with the number of thermal pulses that the star will experience.

Below I plot the C/O-number ratio for the different models.

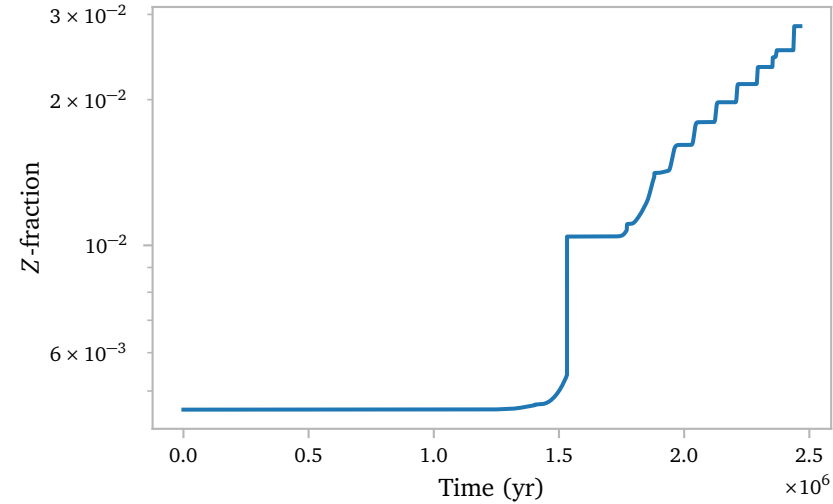


Both the solar model and the $Z = 0.0883$ model look as expected, with the subsolar model achieving a slightly higher C/O-ratio, as expected.

The $Z = 0.00453$ model shows the strong bump that we saw in some of the earlier higher mass models as well.

Since I have a lot of profiles for this star, we can try to work out what is happening¹.

Let me start by plotting the Z-fraction at the surface of all profiles for the low-metallicity model.

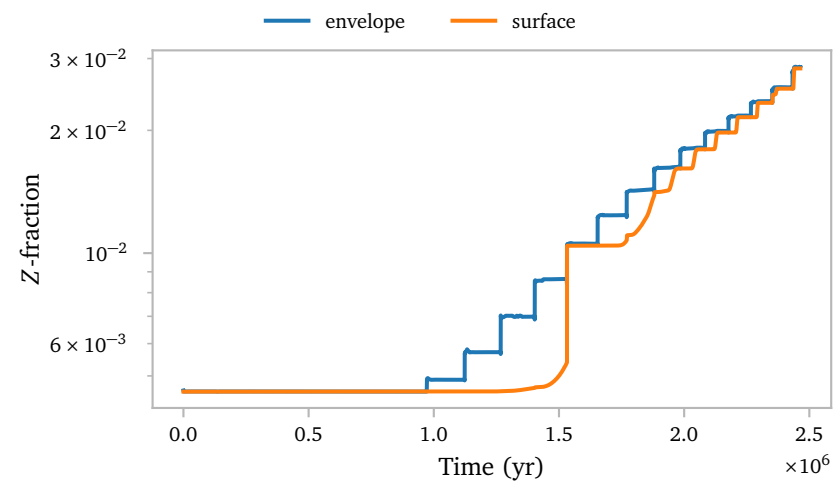


This looks a lot like the plot above, which is good, because that means we can use this Z-fraction as a way to understand the sudden jump.

I really like [this interactive figure](#). It shows:

1. Repeated hydrogen shell burning followed by unstable helium burning.
2. The creation of metals during he-burning followed by dredge up.
3. The loss of the envelope
4. The effect of the superadiabatic sub-surface layers on the metal abundances at the surface.

Clearly the abundance at the surface is not necessarily the abundance in most of the envelope. Below, I try to show that even the model with the sudden jump in surface C/O-ratio still has a well-behaved envelope abundance over time.



The envelope abundance follows a much more predictable staircase-like pattern.

Since the superadiabatic subsurface layers contain very little mass, the envelope value should provide a much better estimate for the actual abundances that get transferred during RLOF and wind mass transfer.

References

Herwig, F. (2005) *Evolution of Asymptotic Giant Branch Stars*. [ARA&A43, 435](#).

Karakas, A. I., Lattanzio, J. C., & Pols, O. R. (2002) *Parameterising the Third Dredge-up in Asymptotic Giant Branch Stars*. [PASA19, 515](#).